Adarsh Sahu

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Summary

Production and Industrial Engineering undergraduate at NIT Jamshedpur applying machine learning, statistics, and operations research to high-impact operational problems — credit risk, transit forecasting, search ranking, and fraud detection — with an emphasis on rigorous evaluation and measurable business return.

Education

National Institute of Technology Jamshedpur 2024 – 2028
B.Tech, Production and Industrial Engineering *Jamshedpur, Jharkhand*
Coursework: Operations Research, Engineering Statistics, Production Management, Engineering Economics.

Experience

OA Helper | Growth Intern Jul 2026 – Present
Remote — Ranchi, Jharkhand
- Built the campus acquisition channel nationwide — student communities, coding clubs and campus representatives — to drive platform adoption across engineering colleges.
- Translated campus feedback into targeting and positioning decisions, working directly with the product team on user-acquisition strategy.

Projects

Collections Intelligence — Credit Recovery Optimizer | *Python, XGBoost, TreeSHAP, SciPy*
- Engineered a dual-propensity classification framework for 20,000 delinquent borrowers, decoupling behavioral **Willingness to Pay** from financial **Ability to Pay** to optimize call-center targeting and reduce recovery budget waste.
- Audited and eliminated 9 data-leakage variables; benchmarked models using global banking risk standards: **ROC-AUC 0.7226**, **Gini Coefficient 0.4451**, and **Kolmogorov-Smirnov (KS) Statistic 34.15%**.
- Applied **TreeSHAP** to interpret feature mechanics: proved Willingness is governed by promise-to-pay adherence and contact responsiveness, while Ability is governed by salary credit frequency and debt-to-income.
- Simulated a 30% contact-capacity constrained call center via a 2×2 decision matrix, delivering an **INR 36.97 Crore net cash recovery uplift** over the 34% random baseline.

RailCross — Railway Crossing Delay Assistant | *Python, scikit-learn, TypeScript, Next.js*
- Predicts railway level-crossing state from traffic and stopped-time signals, on an interactive map.
- Evaluated on a crossing-level holdout — 3 of 14 crossings withheld entirely from training — so figures describe unseen locations: **ROC-AUC 0.888**, **F1 0.593** against 0.16 for a rule baseline.
- A calibrated UNKNOWN abstention band lifts accuracy on decided cases to ∼90%, declining rather than guessing on stale inputs. Synthetic benchmarks only.

PRAHARI — Digital Fraud Intelligence Platform | *JavaScript, Naive Bayes, Rule Engine*
- Client-side scam detector with two independent engines — a weighted-signal rule engine covering 10+ fraud families in English, Hindi and Hinglish, and a character-trigram Naive Bayes classifier.
- Scored by leave-one-out cross-validation over a 48-message corpus whose 24 negatives are real bank OTP and KYC texts, with a threshold slider exposing the precision/recall trade-off. **ET AI Hackathon 2026**.

Search Ranking Model — Learning to Rank | *Python, XGBoost Ranker, BM25, TF-IDF*
- XGBoost ranker trained on the MSLR-WEB10K benchmark (136 features, graded human relevance judgements), with BM25, TF-IDF, freshness and popularity features.
- Evaluated with **NDCG@10** and **MRR** rather than accuracy, since position and graded relevance — not correctness — define a ranked list: **NDCG@10 0.5581**, **MRR 0.8516**.

Also

Projects: Olist E-Commerce SQL Analytics (PostgreSQL joins, CTEs, GMV modeling) • M5 Walmart demand forecasting (Prophet vs XGBoost) • IBM Telco churn prediction (0.84 ROC-AUC) • RAG study assistant (FAISS)
Competitions: XAVENIR 2026 — hallucination detection without pretrained models • MOVES 2026
Leadership: Core Team Member, Phocus NIT Jamshedpur • Volunteer, National Service Scheme

Technical Skills

Languages & Tools: Python, SQL, JavaScript/TypeScript, pandas, scikit-learn, XGBoost, PostgreSQL, FastAPI, Git
Machine Learning & Statistics: Classification, ranking (NDCG/MRR), anomaly detection, credit risk scoring, time-series forecasting, gradient boosting, calibration and abstention, cross-validation and leakage control, hypothesis testing, experiment design
AI Systems: RAG, embeddings, FAISS, TreeSHAP **Strengthening:** PyTorch, Docker, MLflow